PROBLEM & AUDIENCE

Reading reveals interests. Discovery still starts over.

People repeatedly translate what they read into new searches across video, news, and social platforms.

WHO THIS HELPS

Students and researchers

Curious readers

People exploring educational content

Goal Discover relevant, diverse content from the interests expressed in a user’s reading.



Read2Discover: a personalized discovery agent

The agent turns user-provided reading material into an explained, cross-platform feed.



KEY CAPABILITIES

01

Understands the reading

Identifies topics, entities, and related interests.

02

Searches autonomously

Generates source-specific queries and retrieves candidates.

03

Ranks with guardrails

Balances relevance and diversity, then removes duplicates.

04

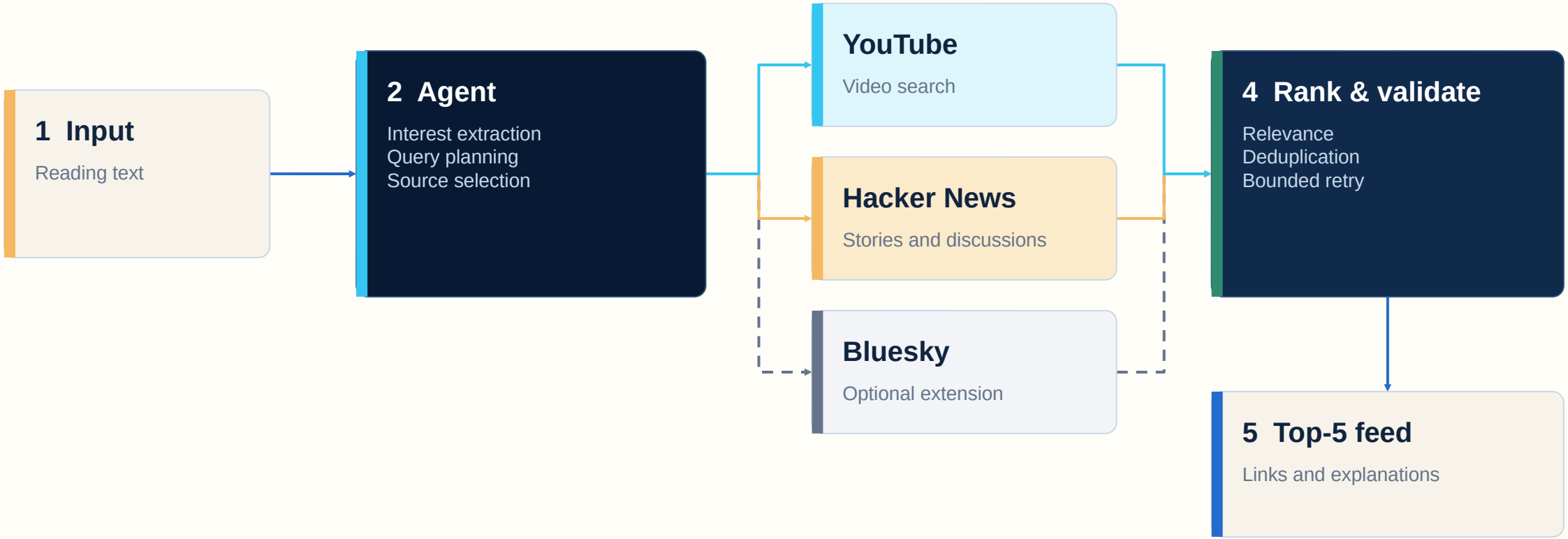
Explains each match

Shows the source link and why the item fits the reader.

Expected output: a Top-5 feed with at least two content formats and a short relevance explanation for every item.

A bounded agentic workflow

The LLM decides what and where to search. Python tools retrieve, filter, and validate results.



TECHNOLOGIES Python · Streamlit · LLM API · YouTube Data API · Hacker News API

Manageable scope and explicit guardrails

The minimum viable product tests the agentic search loop without building recommendation infrastructure.

MINIMUM VIABLE PRODUCT

- Text-based reading input
- YouTube and Hacker News retrieval
- Agent-driven search planning
- Top-5 ranked results with explanations
- Simple Streamlit interface

OPTIONAL

Bluesky integration and refinement from user feedback

OUT OF SCOPE

Model training, authentication, browser history tracking, and large-scale infrastructure

MAIN RISKS AND MITIGATION

Risk	Mitigation
API limits or failures	Cache results and handle source failures gracefully
Irrelevant recommendations	Re-rank candidates and allow one query retry
Duplicate or invalid items	Deduplicate results and validate URLs
Excessive complexity	Limit sources, calls, and retry depth

Scope rule: two required sources, one optional source, one retry.

Four milestones with measurable success criteria

Evaluation compares the agent’s recommendations with a basic keyword-search baseline.

DEVELOPMENT PLAN

01 **Extract interests**

Topic and query generation

02 **Connect sources**

YouTube and Hacker News

03 **Rank and validate**

Diversity, links, bounded retry

04 **Build and test**

Streamlit demo and evaluation

SUCCESS CRITERIA

Metric	Target
End-to-end task completion	**≥ 90%**
Top-5 recommendation relevance	**≥ 80%**
Content diversity	**≥ 2 relevant subtopics**

EVALUATION

Test 10–15 predefined reading scenarios and compare relevance with keyword search.

FINAL DEMO **1 Paste reading material** **2 Agent plans searches** **3 Retrieves and ranks** **4 Personalized feed**